Training a Differentiable Conceptual Functional Equivalent of the US National Water Model to Estimate Parameters for Use in NextGen

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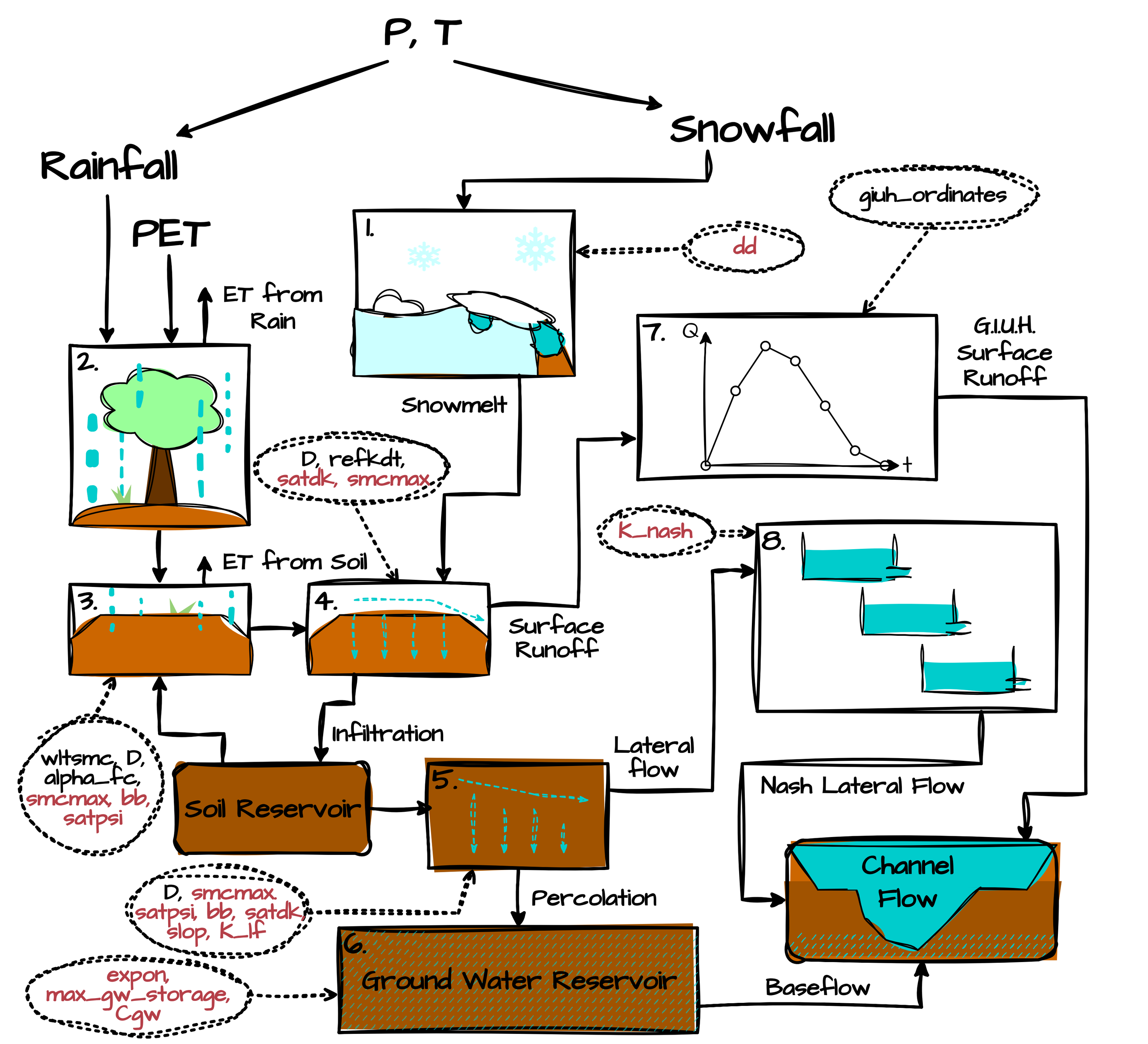
EGU Abstract Link

Background

- NOAA's Next Generation Water Resources Modeling Framework (NextGen) requires parameter estimation but it is challenging especially at locations with insufficient observations to constrain the model.
- The National Water Model Conceptual Functional Equivalent (CFE) is a simplified but functionally equivalent model to the US National Water Model[5] and needs calibrated parameters.
- Differentiable ML has shown potential in hydrologic modeling as a compromise between the pure-ML approach and process based modeling (PBM) approach, here's how it works [7]:

$$\widehat{Q} = g^{\theta}(u, x, A), \tag{1}$$

here, g is the CFE model that depends on parameters $\theta = f^{W}(u, x, A)$, and f is a LSTM with weights found by minimizing a loss function $W = \operatorname{argmin} L(Q, \widehat{Q})$.



1. Snowmelt and snow storage based on degree-day method.
2. Scheme for ET from rain based on PET.
3. Budyko-style soil scheme for ET.
4. Schaake infiltration runoff scheme, adjusted to soil storage.
5. Soil moisture subroutine for percolation and lateral flow, adjusted with soil outflux and groundwater conditions.
6. Nonlinear groundwater subroutine for baseflow.
7. Convolutional G.I.U.H. routing subroutine for surface runoff.
8. Nash cascade subroutine for lateral flow.

Figure 1. Diagram of CFE with a simple snow module attached, courtesy of Pulling et al [6]. All parameters estimated by the framework are in red.

- Building on previous efforts [2], we developed a fully differentiable CFE model. We validated it against the original CFE model [5]. We then added a snow module and embedded the model within the NeuralHydrology platform [4], producing a coupled framework that we call NH-dCFE.

We investigated NH-dCFE's potential for parameter regionalization. We introduce two ways of training static parameters, validate these methods' robustness on SHM [1], and compare their performance to a LSTM benchmark and a previous effort.

Training & Parameters

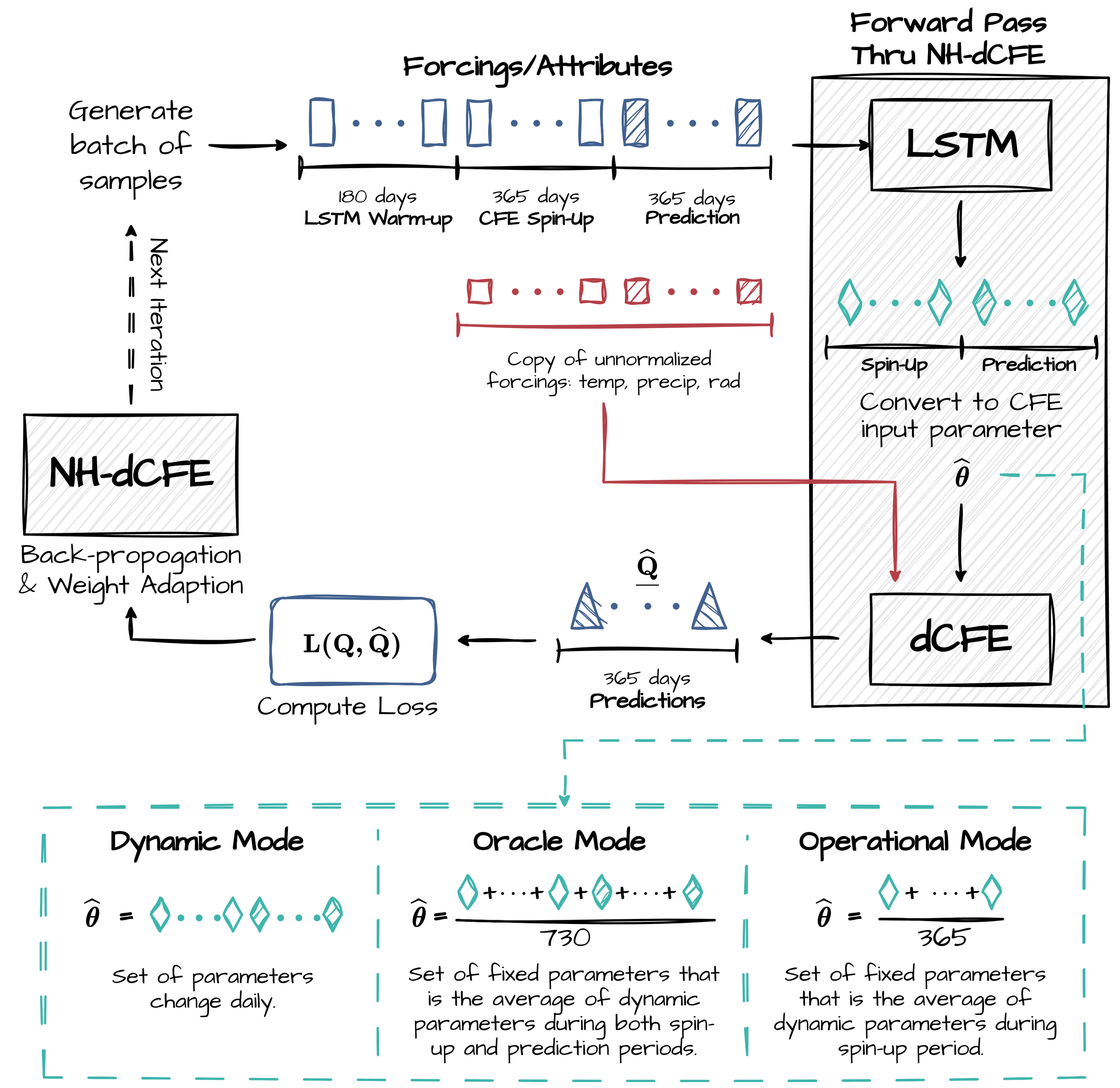


Figure 2. Diagram of each iteration of the training process. Training set-up: 1/2 years LSTM spin-up, 1 years CFE spin-up after, and then 1 year daily outputs for back-propagation.

Pure-LSTM vs. Hybrid Performance

Assessing performance of dynamic modes for dCFE and SHM compared to pure-LSTM given approximately the same training settings.

The same:

- Number of basins: 516; Epoch: 30; Loss: MSE
- Training dates: 01/10/2000 – 30/09/2014
- Validation dates: 01/10/1990 – 30/09/1996
- Testing dates: 01/10/1998 – 30/09/2002
- Dynamic inputs & static inputs, etc...

Unique for NH-dCFE & SHM:

- LSTM warm-up: 180
- Conceptual spin-up: 365
- Sequence length: 910

Unique for pure-LSTM:

- Sequence length: 365

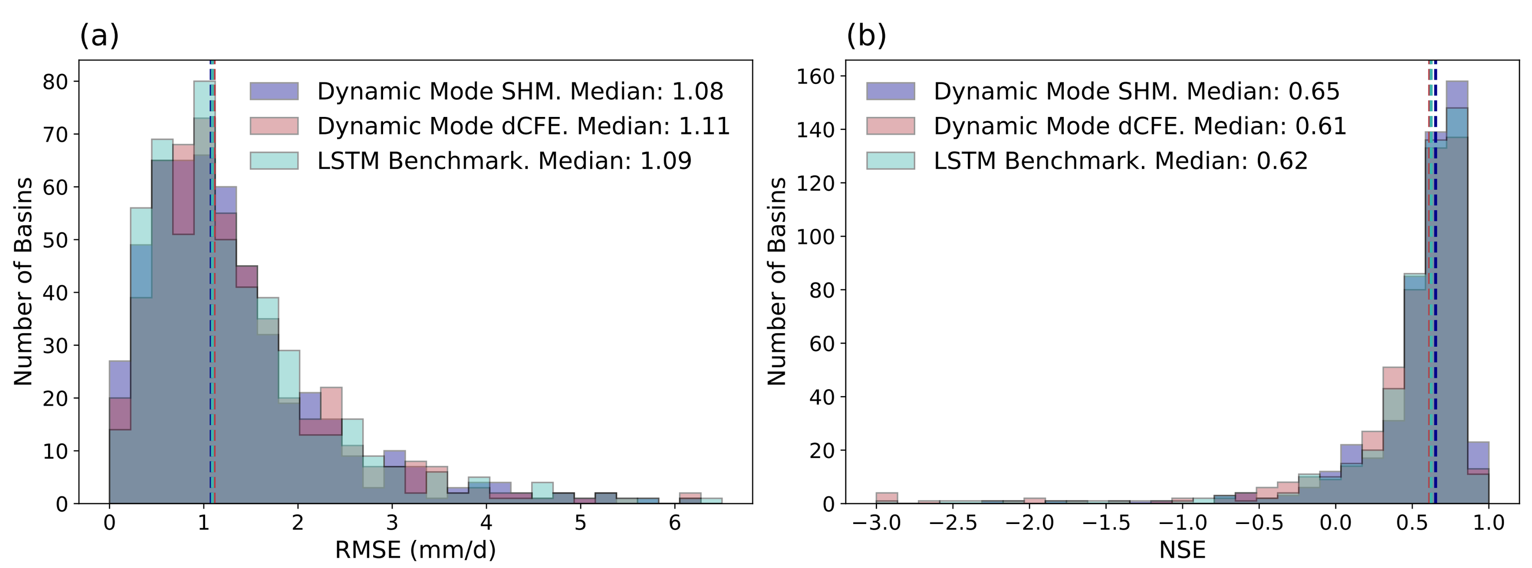


Figure 3. Histograms of per-basin metrics on the testing set for pure-LSTM benchmark, SHM and dCFE in dynamic mode, vertical lines mark the medians. (a) shows that for RMSE; (b) shows that for NSE.

Hybrid models perform similarly if not better than pure-LSTM in this example.

Dynamic vs. Static Parameters Performance

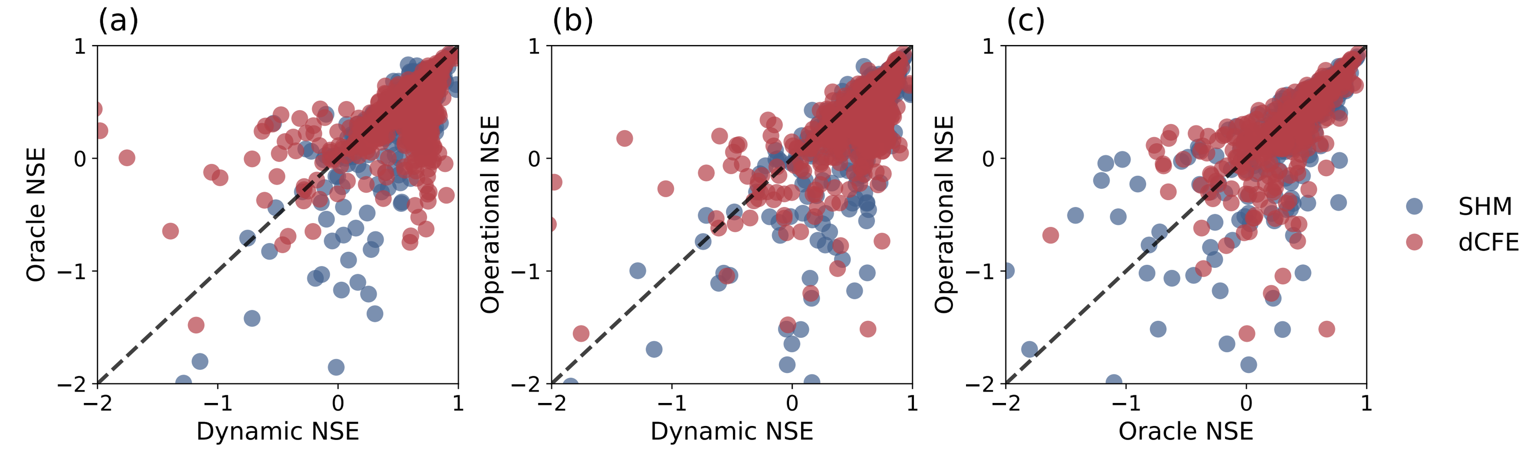


Figure 4. Scatter plots of NSEs from different parameter modes over the testing set for SHM (in blue) and dCFE (in red). (a) shows that for dynamic vs. oracle modes; (b) shows that for dynamic vs. operational modes; (c) shows that for oracle vs. operational modes.

Dynamic mode performs the best. The operational mode is almost as good as oracle mode. SHM sees a larger drop in performance when using static parameters than dCFE.

Comparison to A Previous Effort

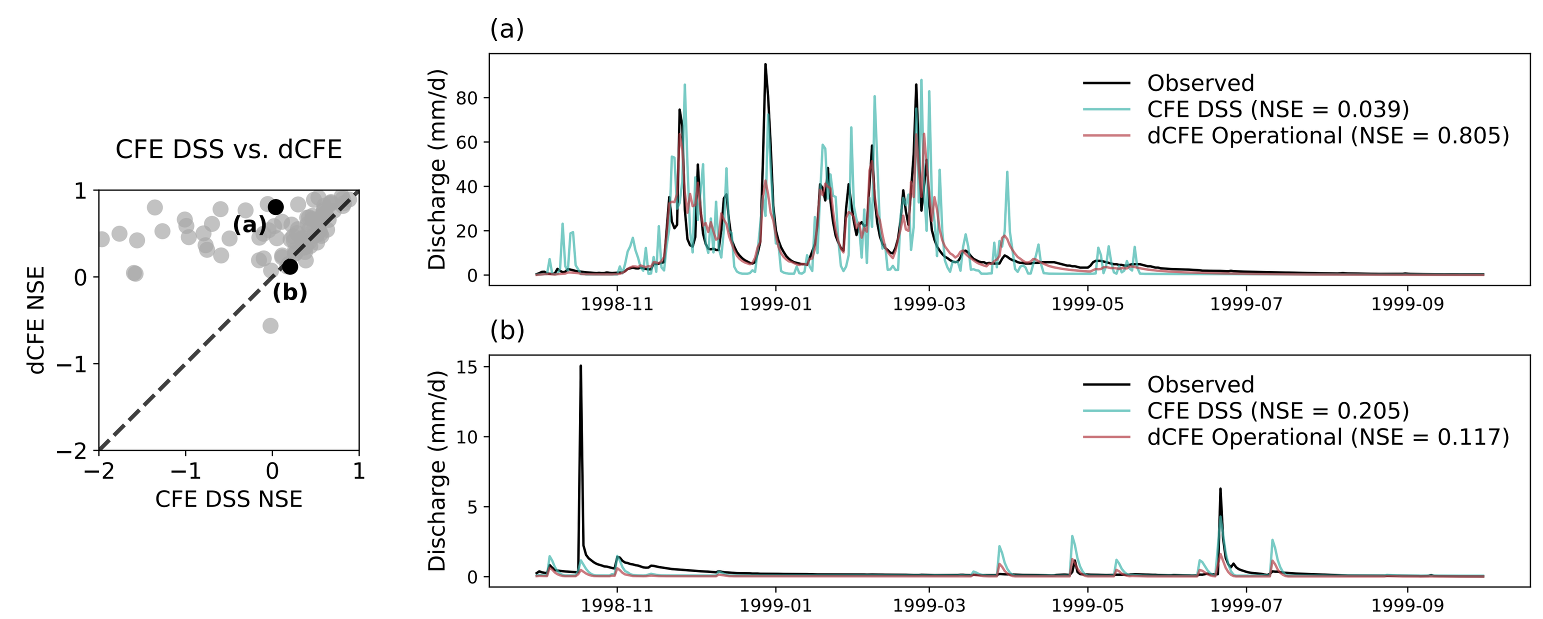


Figure 5. Left: scatter plot of NSE results using parameters calibrated by Dynamically Dimensioned Search (DSS) Optimization Algorithm from Araki et al [3], against NH-dCFE (without snow module) operational mode results. Right: (a) shows 1 year of runoff for Tucca Creek near Blaine, OR (ID: 14303200); (b) shows 1 year of runoff for Dry Frio River near Reagan Wells, TX (ID: 08196000).

Parameters & Internal States

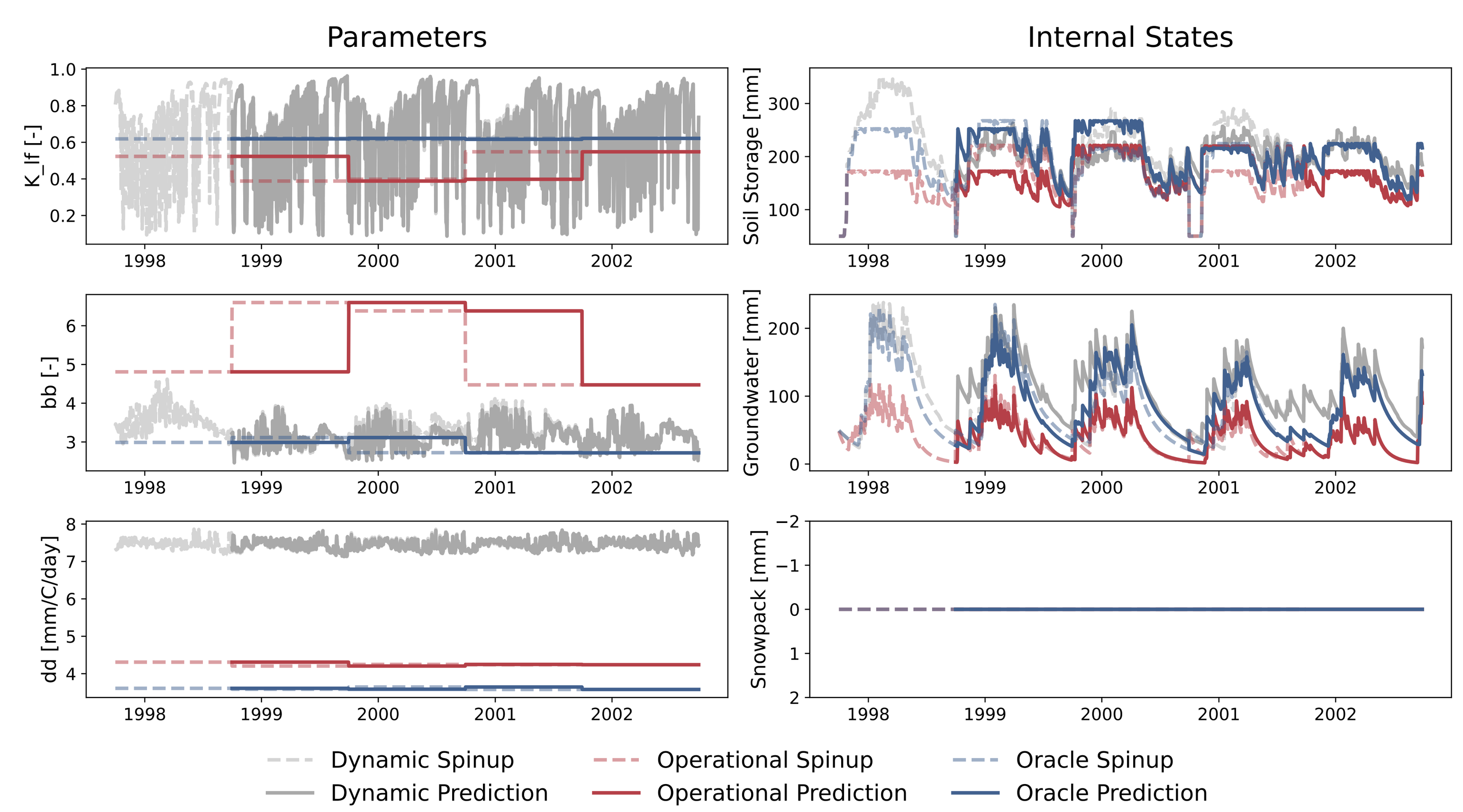


Figure 6. Plots from trained dCFE for Chattooga River near Clayton, GA (ID: 02177000). Left: three plots showing CFE parameters, from possibly identifiable, semi-identifiable, and not identifiable. Right: internal states possibly heavily impacted by parameters on the left accordingly.

Summary

Successes:

- Integrated dCFE into NeuralHydrology.
- Demonstrated viability of differentiable ML for static parameter estimation.
- Created user-friendly way to examine parameters and internal states.
- Extended the SHM structure in the NH modelzoo to facilitate training more complex dPBMs.

Challenges:

- Long training times for hourly models.
- Exploding or vanishing gradient in static parameter modes that require care.
- Interpretability and identifiability of parameters being estimated.
- Long warm-up and spin-up periods decrease amount of data for training, validation, and testing.
- Scalability.

Future Directions

- Ablation study to test the sensitivity of each CFE parameter and examining Fisher Information eigenvalues to determine their identifiability.
- Refine code and provide resource to community for using NeuralHydrology for differentiable modeling.
- Resolve computational issues to enable hourly training.
- Assess performance on ungauged basins and scalability to NextGen.

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